AP CALCULUS AB

Unit 1 • Unit 1 Pretest • Limits, Continuity, and Differentiability

Unit 1 Pretest • Limits, Continuity, and Differentiability

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Part A • Multiple choice

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$ is

(A) 0

(B) 3

(C) 6

(D) nonexistent

Question 2

NO CALCULATOR

$\lim_{x \to \infty} \dfrac{5x^2 - x}{2x^2 + 3}$ is

(A) 0

(B) $\frac{5}{2}$

(C) 5

(D) ∞

Question 3

NO CALCULATOR

$\lim_{x \to 1^-} \dfrac{2}{x - 1}$ is

(A) ∞

(B) 2

(C) 0

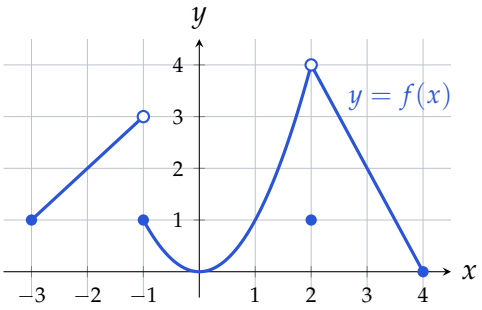
(D) $-\infty$

Question 4

NO CALCULATOR

$\lim_{x\to 0} \dfrac{\sin 3x}{x}$ is

(A) 0
(B) 1
(C) 3
(D) $\frac{1}{3}$



Question 5

NO CALCULATOR

Using the graph of f, which limit does not exist?

(A) $\lim_{x\to 0} f(x)$
(B) $\lim_{x\to -1} f(x)$
(C) $\lim_{x\to 2} f(x)$
(D) $\lim_{x\to 3} f(x)$

Question 6

NO CALCULATOR

Using the graph of f, at which x is the discontinuity removable?

(A) $x = -1$
(B) $x = 0$
(C) $x = 2$
(D) $x = 3$

Question 7

NO CALCULATOR

For which k is $f(x) = \begin{cases} kx^2, & x \le 1 \\ x+3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

Question 8

NO CALCULATOR

f is continuous on $[2,6]$, $f(2)=5$, $f(6)=-3$. Which must be true?

(A) $f(4)=1$
(B) $f(c)=0$ for some c
(C) f has no maximum
(D) $f'(c)=-2$

Question 9

NO CALCULATOR

Which describes $\lim_{h\to 0}\dfrac{(2+h)^3-8}{h}$?

(A) $f'(2)$ for $f(x)=x^3$, equal to 12
(B) $f'(8)$, equal to 3
(C) 0
(D) 8

Question 10

NO CALCULATOR

$f(x)=|x+1|$ is not differentiable at

(A) $x=1$
(B) $x=0$
(C) $x=-1$
(D) nowhere

Part B • Free response

Question 11 (Free response)

NO CALCULATOR

Let $g(x)=\begin{cases}\dfrac{x^2+x-6}{x-2}, & x<2\\ cx+1, & x\geq 2.\end{cases}$

(a) Find $\lim_{x\to 2^-} g(x)$.
(b) Find c so that g is continuous at $x=2$; justify with the definition.
(c) With this c, is g differentiable at $x=2$? (The left piece simplifies to $x+3$.)

Question 12 (Free response)

NO CALCULATOR

A differentiable function W (temperature, °C) has $W(0)=20$, $W(3)=24$, $W(5)=31$, $W(8)=27$.

(a) Must $W(t)=30$ for some t in $(3,5)$? Justify.
(b) Must $W'(t)=3.5$ for some t in $(3,5)$? Justify.
(c) Estimate $W'(4)$, with units.